CS230 – REQUIREMENTS ENGINEERING

Software Requirements Specification

Student Administration System

Group 12

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1.0 Introduction

1.1 Purpose

This Software Requirement Document aims to clarify complete software requirements, including functional and non-functional requirements, and provides a clear, intuitive presentation of the Student Administration System (SAS).

This document is intended for stakeholders including the Project Team, Staff of the School: Faculty members and Administrative Staff, Students and Parents.

1.2 Scope

The Student Administration System is developed to facilitate communication among students, parents and educators, improving student information management and supporting decision making. The system accomplished the above goals by providing student information repository, student reports generation and academic monitoring system, attendance and discipline tracker, timetable generation and scheduling, student activities tracker, communication and parental engagement system, staff management and library resources management. Data security and student privacy are also included in the scope of this system

1.3 Definitions

SRS – Software Requirements Specification: an overall description of the software system to be developed, including functional and non-functional requirements of the system, interactions between users and the system.

SAS – Student Administration System: The system is to be developed in this project.

DBMS – Database Management System: a software used to manage data from a database.

HTTPS - Hypertext Transfer Protocol Secure: an advanced version of HTTP which is an application layer protocol using encryption to protect data communication over a computer network.

SSL – Secure Socket Layer: an encryption protocol which protects Internet communication

AWS EC2 - Amazon Web Services Elastic Cloud Compute: a cloud computing service provided by Amazon.

AWS S3 - Amazon Simple Storage Service: a cloud-based object storage service provided by Amazon.

MFA - Multifactor Authentication: an electronic authentication method which allows users to access an account only if they provide two or more evidence

RBAC - Role-Based Access Control: a method to restrict system access to authorized users based on their roles.

KISS – Keep It Simple, Stupid: a design principle that emphasizes the simplicity of design/system

CI/CD – Continuous Integration/Continuous Delivery: the combined software development principle, referring to frequent merging of several changes into main branch and automatic new software functionality releasement.

UI – User Interface: the elements of a website or app that the user interacts with.

UTC - Coordinated Universal Time: the primary time standard used to regulate clocks and time in the world.

UTC-12: a time zone that is 12 hours behind UTC, referring to Fiji time in this project.

ARIA – Accessible Rich Internet Application: a way to improve the accessibility of web content and applications for users with disabilities.

Lazy Loading: a technique to defer loading of an object until it is needed.

UAT – User Acceptance Testing: a phase in software development which invites end-users to test the software in real-world scenario to ensure the software meets excepted requirements.

GDRP - General Data Protection Regulation: an European Union regulation on information privacy and security.

ROI - Return on Investment: a financial metric presents the comparison between investment's gain and its cost.

API - Application Programming Interface: a connection between computers or computer programs.

REST API - Representational State Transfer API: a type of API that follows the design principles of the REST architectural style which is used to connect distributed hypermedia systems.

1.4 References

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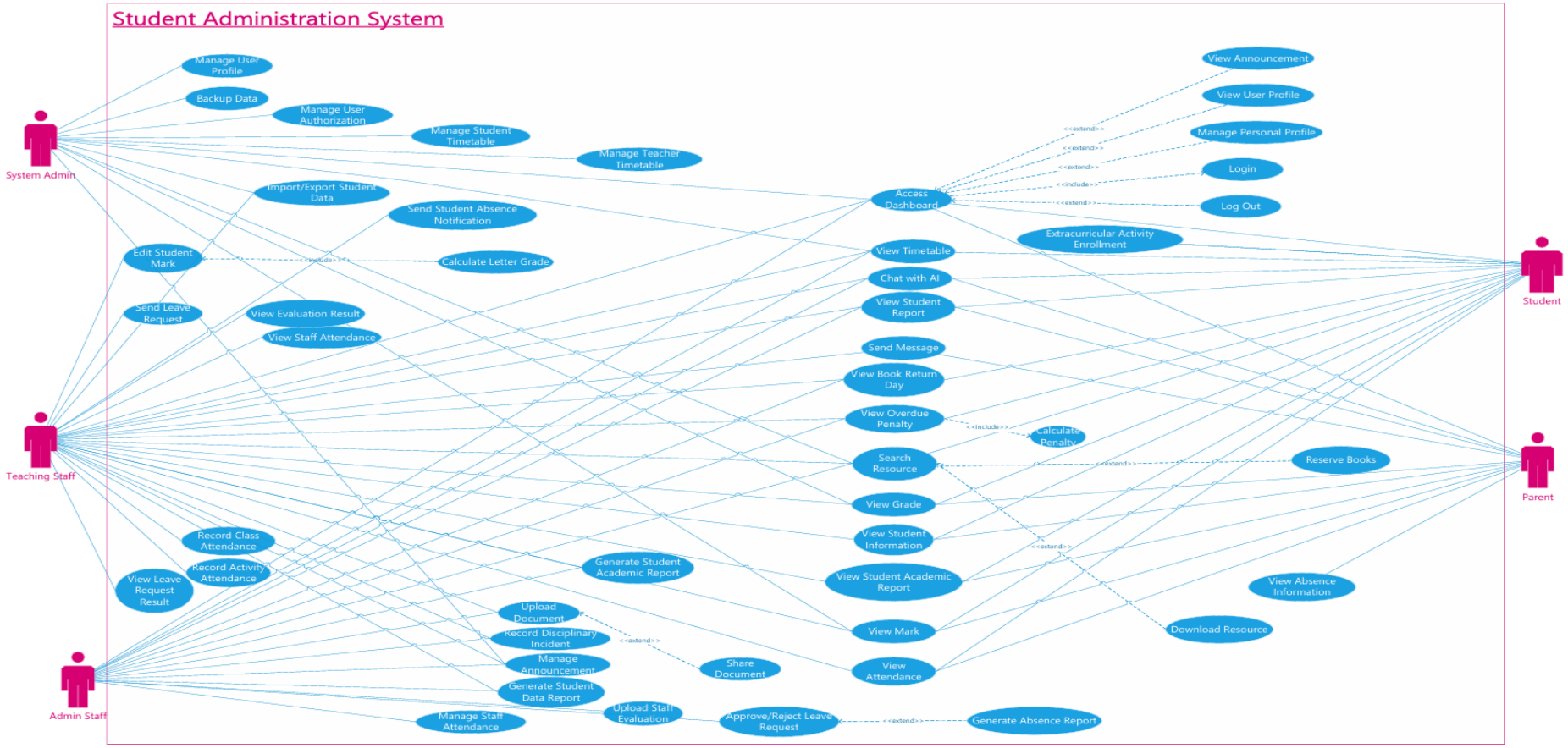
2.0 Overall Description

2.1 Systems Environment



2.1.1 Description

The above diagram represents a visualization of the actors, which means users, and the use cases, which refers to interaction between users and the system. This system includes 5 actors: System Administrator, School Staff which includes Teaching Staff and Administring Staff, Student and Parents. The system administrator has the highest authority of the system, and other users have different operating authorities depending on their roles.

2.2 Operating Environment

2.2.1 Required Software

The software specifications required to run the proposed system on mobile and desktop devices are as follows:

2.2.1.1 Mobile

In order for the proposed system to operate smoothly, without crashing on mobile devices, it needs any one the following operating systems

- Android OS
- Apple iOS

2.2.1.2 Desktop

One of the following operating systems is required for the proposed system to operate consistently on desktop devices

- Windows
- macOS

2.2.2 Required Hardware

The hardware required to run the proposed system on mobile and desktop devices are as follows:

2.2.2.1 Mobile Devices

The following are the minimum hardware requirements to operate the proposed system on Android and iOS devices

i. For Android devices
    - Android 11 or above
    - 2GB of minimum ram
    - 4GB of minimum storage

ii. For iOS devices
    - iOS 13 or above
    - 2GB of minimum ram
    - 4GB of minimum storage

2.2.2.2 Desktop

The following are the minimum hardware requirements to operate the proposed system on Windows and macOS devices

i. For Windows devices
   - Windows 8.1 or above
   - 4GB of minimum ram
   - 6BG of minimum storage
   - Intel i5 processor or above

ii. For macOS devices
   - macOS 10.12 or above
   - 4GB of minimum ram
   - 6GB of minimum storage
   - Apple M1 chip or above

2.3 Constraints

2.3.1 Implementation and Maintenance Cost

The cost of implementing and maintaining the student administration system in the real world can become very high, making it a constraint. The high implementation cost results from the investment in cloud services for server hosting and database management, providing software improvements and constant security patches through updates and utility costs such as internet and electricity as part of system usage.

2.3.2 Network Access

The student administration system heavily depends on internet connectivity in order to perform its day-to-day operations. Network access issues such as low bandwidth or disrupted network access (mostly for rural areas) may result in system downtime. This will become a barrier to the users, disrupting operations such as taking attendance, uploading grades and scheduling tasks.

2.3.3 Resource and Programming Language

The availability of the resources that are present during the development of the student administration system imposed major constraints for the student administration system. A constraint imposed upon the development of the system was the requirement for it to be built using Node.JS and Java, creating a major constraint in terms of the programming language to be used as the choice of the programming language was limited. Functionality and availability of other tools such as capacity of the server and capabilities of the network resulted in constraint in terms of resources available to test and deploy the system.

Managing these resources available to the development team ensured a successful completion of the proposed student administration system.

2.3.4 User Training

The user training becomes a constraint in terms of making training materials such as guides, conducting training sessions, that is hiring a professional trainer and choosing an optimum venue for the training session to be conducted and the trainer's time if the same training needs to be conducted repeatedly due to the inability of the trainees to understand the operation of the student administration system.

2.4 Assumptions and Dependencies

2.4.1 Assumptions

2.4.1.1 Network Connectivity

It is assumed that the devices used to operate the student administration system will be accessible to a network connection that is reliable. The devices may use internet provided by the local internet service providers such as Vodafone, Digicel, Telecom Fiji Limited and Starlink, which is available globally.

2.4.1.2 Operating System Availability

It is assumed that the operating system being used for the student administration system will be available with the purchase of the designated desktop and mobile devices. The operating systems being used for the proposed system are Microsoft Windows, macOS, Android and Apple iOS.

2.4.1.3 Operational Skills

It is assumed that the people using the student administration system will have basic knowledge in terms of using a computer system to some extent, together with having some sort of knowledge on day-to-day programs being used such as Microsoft Word and Microsoft Excel. Having such operational skills will ensure that less user training is required.

2.4.2 Dependencies

2.4.2.1 Database System

The student administration system depends on the PostgreSQL for database and DynamoDB for database management. This will eliminate the need to pay licensing fees, which will reduce upfront costs.

2.4.2.2 Storage

The student administration system depends on two main types of storage. These include SSDs as part of physical storage and iCloud and OneDrive as part of cloud storages. The physical storages may come in handy if the users wish to download

files into their devices while the cloud storages may come in handy if the users wish to upload files from their devices.

2.4.2.3 User Creation

The student administration system depends on the system admin to create the respective user accounts into the system. This includes the user's login credentials such as the username, password and the user's email address.

2.5 Stakeholders of the system

The proposed student administration system has two categories of stakeholders. These are internal and external stakeholders.

2.5.1 Internal Stakeholders

Internal stakeholders involve people who are within the organisation and have a direct interest from the system

2.5.1.1 School Administration

The school administration consists of the school Principal, Vice Principal, Assistant Principal and the board management members of school. The school administration oversees the day-to-day operation of the school system and create tactical and strategic plans for the long-term goals of the school.

2.5.1.2 Teaching Staff

The academic staff includes teaching staff who may be respective department heads or subject teachers. Some of the roles of the academic staff includes creating lesson plans, assessments, recording class attendance and other general tasks.

2.5.1.3 Students

This consists of the students who are enrolled in the school and are attending their respective classes. Some of the roles of the students in the school are taking part in academic activates and optional participation in extracurricular activities such as sports.

2.5.2 External Stakeholders

External stakeholders include people outside the organisation having either a direct or an indirect interest from the system.

2.5.2.1 Parents and Guardians

Parents and guardians are people whole are known as primary caregivers to students. Parents and guardians also oversee their child(ren)'s performance in school and provide the needed financial, academic and emotional support. The

parents and guardians are also the people who fill consent forms and attend parent-teacher meetings every term.

2.5.2.2 Educational Authorities

Educational authorities refer to the Ministry of Education, which is the parent body of all academic institutes in the country. The Ministry of Education authorizes the standards and the policies that govern the academic institutes. This involves setting the yearly overall outlines for example, external exam schedules and school holidays.

2.5.2.3 Higher Education Institutions

The higher education institutions include educational bodies on the tertiary level such as Universities. Universities may require certain documents such as birth certificate from students during the enrolment process. These types of documents are easily available in the student profile which is made by the school that the student attends.

3.0 Requirements Specification

3.1 Detailed functional requirements

3.1.1 Create New User Profile

FR1: The system should allow the system admin to create new user profile

Title:	Create New User Profile
Description:	System Administrators can create new user accounts with appropriate roles and permissions. Creations consist of new user accounts intended for new users of the system.
Actor(s) Involved:	System Administrator
Preconditions:	• System Administrator must be logged in to the system. • System Administrator should have the required permissions under RBAC to create new user profiles.
Post-Conditions:	• A new user profile is created in the system database. • User is given the appropriate roles and permissions. • The system logs the new account creation.
Triggers:	• The system administrator selects the “Add New User” option from the User/Account Management interface.
Activity Flow:	1. The system admin navigates to the User/Account Management section. 2. The System Administrator then selects the “Add New User” option. 3. System then displays the account creation form. 4. The System Administrator enters the required fields of user information (name, email, role, etc). 5. The System Administrator then submits the form. 6. System then validates the user information in the submitted form. 7. System then creates the new user profile. 8. System displays a success message confirming the account creation.

Exception Conditions:	• When required fields are missing (System will then give a prompt message to fill in the mandatory fields). • When the email entered may already be in use - pre-existing user or account email. • Invalid data formats of input will display an error message.

3.1.2 Update User Profile

FR2: The system should allow the system admin to update user profile

Title:	Update User Profile
Description:	System Administrators can modify existing user account information such as personal information, contact details, user roles and assigned permissions.
Actor(s) Involved:	System Administrator
Preconditions:	• System Administrator must be logged in to the system. • The System Administrator should have the necessary permissions to be able to update existing user profiles. • The user profile that is to be updated should be existing within the system.
Post-Conditions:	• User profile involved will be updated with new information details. • The system logs the updates made to the user profile.
Triggers:	• The System Administrator selects the “Edit Profile” option for the user profile that is to be updated.
Activity Flow:	1. The System Administrator navigates to the User/Account Management section. 2. The System Administrator locates the desired user profile. 3. The System Administrator selects the “Edit Profile” option. 4. System displays the user profile with the current information. 5. The System Administrator modifies the relevant fields that are meant to be updated. 6. The System Administrator then confirms these changes by selecting the “Save All Changes” option. 7. System validates the updated information. 8. System updates the user profile. 9. System displays a success message confirming the account creation.

Exception Conditions:	• Invalid data formats used will display an error message. • Attempting to edit information on a deleted account will prompt the system to display an error message.

3.1.3 Delete User Profile

FR3: The system should allow the system admin to delete user profile

Title:	Delete User Profile
Description:	System Administrators can remove user accounts from the system database when they are no longer needed.
Actor(s) Involved:	System Administrator
Preconditions:	• System Administrator must be logged in to the system. • The System Administrator should have the necessary permissions to be able to delete existing user profiles. • The user profile that is to be deleted should be existing within the system.
Post-Conditions:	• The deleted user profile is removed from the list of active users. • The system maintains an archive of the deleted user account for audit purposes. • The system logs the deleted user account.
Triggers:	• The System Administrator selects the “Delete Profile” option for the user profile that is to be deleted.
Activity Flow:	1. The System Administrator navigates to the User/Account Management section. 2. The System Administrator locates the desired user profile. 3. The System Administrator selects the “Delete Profile” option. 4. System prompts for confirmation to delete profile. 5. The System Administrator confirms to delete the profile. 6. System removes the user from the list of active users. 7. System archives the user data record. 8. System displays a success message confirming the account deletion.
Exception Conditions:	• User has active dependencies (assigned roles) - system will display a message to the System Admin. • System error during deletion (If the user is the only system administrator, then deletion is prevented) - system will display an error message.

3.1.4 Perform Data Backup

FR4: The system should allow the system admin to perform the data backup process

Title:	Perform Data Backup
Description:	System performs data backup processes of all system data which are manually set up by the System Administrators.
Actor(s) Involved:	System Administrator
Preconditions:	• The storage for backed up data is properly configured. • Sufficient resources are available to perform the data backup.
Post-Conditions:	• A complete backup is created and securely stored. • The backed-up data is verified for integrity reasons. • System creates a log for the backup operation. • System Administrators are notified about the backup status (Pending, Percentage completed or done).
Triggers:	• Manual command by the System Administrator to start the backup process (if manually).
Activity Flow:	1. System initiates the backup process at the command of the System Administrator. 2. System creates a backup of all critical databases and files. 3. System verifies the backup integrity. 4. System compresses and encrypts the backed-up data. 5. System transfers the backup data to the designated storage location. 6. System creates a log of the backup details (time taken, backup size and backup status). 7. System sends a notification message to the System Administrator regarding the backup status.
Exception Conditions:	• Insufficient storage space to store backup files - system will then send an alert message to the Admin.

	• Failure in backup verification - system automatically makes a second attempt to create a backup and also notify the System Administrator. • Network errors - the error is logged by the system and an alert is sent to the System Administrator.

3.1.5 Manage User Roles

FR5: The system should allow the system admin to manage user roles

Title:	Manage User Roles
Description:	System Administrators are able to manage (Assign and Modify) the RBAC (Role-Based Access Control) permissions for other users.
Actor(s) Involved:	System Administrator
Preconditions:	• The System Administrator must be logged into the system. • The System Administrator should have the necessary permissions to be able to manage user roles. • The user profile for which the role is intended to be managed must exist in the system.
Post-Conditions:	• User permissions and roles are updated/modified. • User access should be able to reflect the new role that is assigned to them. • The system will create a log for the user role modification action that has been carried out.
Triggers:	• The System Administrator selects the “Edit User Role” option for the user profile that is to be modified.
Activity Flow:	1. The System Administrator navigates to the User/Account Management section. 2. The System Administrator locates the desired user profile. 3. The System Administrator selects the “Edit User Role” option. 4. System displays the current roles assigned to the user selected and also the current roles available for selection. 5. The System Administrator modifies the roles and assigns them as intended. 6. The System Administrator then confirms these changes by selecting the “Save Changes” option. 7. System updates the user permissions according to the role changes and modifications carried out.

	8. System displays a message confirming the role modification.
Exception Conditions:	• System error during a role update will prompt the system to display an error message. • If an attempt is made to remove all administrative roles, the system will prevent this action.

3.1.6 User Login

FR6: The system should allow the users to login

Title:	User Login
Description:	System will authenticate the users and provide access to the authorized system functionality based on the user credentials and their assigned roles within the system.
Actor(s) Involved:	Teachers, Students, Administrators
Preconditions:	• The user account must be active (not deleted). • Users must have a pre-existing account in the system.
Post-Conditions:	• The user is authenticated and granted system access. • System establishes a secure session for the user. • System creates a log for the successful login.
Triggers:	• User submits the login credentials on the login page.
Activity Flow:	1. The user navigates to the User Login section. 2. User enters the login credentials (username, email & password). 3. User submits the login credentials. 4. System validates login credentials. 5. System verifies the account status. 6. System establishes the user session with the appropriate permissions. 7. System redirects the user to the system dashboard.
Exception Conditions:	• Attempting to log in to the system without filling all required fields will display an error message. • Entering incorrect credentials will prompt the system to display an error message and ask the user to re-enter the correct credentials.

3.1.7 View User Profiles

FR7: The system should allow each user to view other user profiles

Title:	View User Profile
Description:	All authenticated users will have the permission to view the profiles of other users with appropriate privacy controls used which is based on the viewer's role and their relationship to the profile owner.
Actor(s) Involved:	Administrative and Teaching Staff, Students, Parents and the System Administrators (All authenticated users)
Preconditions:	• Users must be authenticated in the system. • The requested profile to be viewed should exist in the system.
Post-Conditions:	• User views the profile information of the intended user based on their permission level.
Triggers:	• User selects the "View Profile" on the profile to be viewed.
Activity Flow:	1. User navigates to the search profile feature. 2. User searches for the specific user profile to view. 3. System determines the viewing permissions based off the viewer's role. 4. System displays the user profile with the appropriate level of details related to the user role. 5. System provides navigation back to the previous screen.
Exception Conditions:	• The searched profile may not exist in the system, if so, the system displays an error message notifying the user. • Profile access may be restricted, if so, the system will display an appropriate message notifying the user.

3.1.8 Edit User Profile

FR8: The system should allow each user to edit his/her profile

Title:	Edit User Profile
Description:	Authorized users are given permission to modify their own profile information and allow administrators to modify any user profile with appropriate permission controls.
Actor(s) Involved:	System Administrators (for editing any profile) Administrative, Teaching Staff and Students (for own profile)
Preconditions:	• Users must be authenticated into the system. • User has the permission to edit the profile. • The profile to be edited must be existing in the system.
Post-Conditions:	• Profile information is updated in the system. • The system logs the profile update action.
Triggers:	• User selects the “Edit Profile” action on the profile to be edited.
Activity Flow:	1. User navigates to the “View Profile” feature. 2. User selects the “Edit Profile” option. 3. System displays the editable profile form. 4. User can modify the fields they are allowed to change under their user permissions (not all fields). 5. The user then confirms these changes by selecting the “Save Changes” option. 6. System validates the information submitted by the user. 7. System updates the profile data using the submitted data. 8. System displays a message confirming the successful update.
Exception Conditions:	• Attempting to submit changes without filling all required fields will display an error message. • Invalid data formats used will display an error message. • The system will preserve the original values entered if there are insufficient permissions for specific fields.

3.1.9 User Logout

FR9: The system should allow user to log out

Title:	User Logout
Description:	All users will have the option to be able to securely terminate their active session.
Actor(s) Involved:	Administrative and Teaching Staff, Students
Preconditions:	• The user must be logged into the system.
Post-Conditions:	• The user session is terminated. • User is redirected to the login page. • System creates a log of the successful logout event.
Triggers:	• User selects the “Log Out” option
Activity Flow:	1. User selects the “Log Out” option from the navigation menu. 2. System the prompts the user for confirmation. 3. System terminates the user’s current active session. 4. System clears the session data. 5. System redirects the user to the login page.
Exception Conditions:	• Session may already have expired, if so, then the system redirects the user to the login page. • System error occurs during the log out event, if so, the system attempts to force the session termination.

3.1.10 Record Student Medical Information

FR10: The system should allow the teaching staff to record student medical information

Title:	Record Student Medical Information.
Description:	Student Medical Information such as health conditions, allergies and medications are securely stored and managed with appropriate privacy controls for the information kept.
Actor(s) Involved:	Teaching Staff
Preconditions:	• The teacher editing the information must be logged into the system. • The user must have the appropriate permissions to be able to access the medical data of students.
Post-Conditions:	• Updated/new medical information is recorded and accessible to the authorized users. • System creates a log of the successful change to the stored medical data.
Triggers:	• New student registration. • Request to update medical information. • Teaching staff accesses the stored medical records.
Activity Flow:	1. User navigates to the student profile section. 2. User selects the “Medical Information” section. 3. System validates the access permissions of the user. 4. System displays the Medical Information form. 5. User then enters or updates the medical data. 6. User then submits the changes made. 7. System now encrypts and securely stores the sensitive medical data. 8. System displays a message confirming the successful changes made to the medical information records.

Exception Conditions:	• The system will restrict the user access if the user trying to update/enter the medical information has insufficient permissions. • In the case of emergency access to information, the system will also provide override procedures to access the information stored.

3.1.11 Import Student Data

FR11: The system should allow the system admin to import student data

Title:	Import Student Data
Description:	Student Data import systems are present in the system to import data from external sources such as obtaining records from other schools (for new students switching schools) and using bio-data forms.
Actor(s) Involved:	System Administrators
Preconditions:	• System Administrator must be logged in to the system. • The import data should be in a compatible data format. • The system should have sufficient resources for the import.
Post-Conditions:	• The student data is successfully imported into the system. • System validates and normalizes the imported data. • System creates a log for the import operation. • System generates an import status and notifies the System Administrator.
Triggers:	• User chooses the “Import Data” option under the Data Management section.
Activity Flow:	1. User navigates to the “Data Management” section. 2. User selects the “Import Data” option. 3. System displays the import interface. 4. User selects the file to be imported and the configuration options for the import process. 5. User initiates the import process. 6. System validates the file format and file content. 7. System performs the data import process. 8. System displays a message confirming the successful import process of the student data.
Exception Conditions:	• Invalid data formats imported will display an error message.

	• Should the data be a duplicate of existing data, the system will provide conflict resolution options such as "Replace/Skip" • If there are any data validation errors, then the system will provide a detailed error message.

3.1.12 Export Student Data

FR12: The system should allow the system admin to export student data

Title:	Export Student Data
Description:	Student Data export systems are present in the system to export student data for students changing schools or for generating academic transcripts for students that are graduating.
Actor(s) Involved:	System Administrator
Preconditions:	• System Administrator must be logged in to the system. • The system should have a defined data export criteria (data type, format, encoding and file paths)
Post-Conditions:	• Student Data is exported in the requested format. • System creates a log for the data export process. • System generates a data export status and notifies the System Administrator.
Triggers:	• User chooses the “Export Data” option under the Data Management section.
Activity Flow:	1. User navigates to the “Data Management” section. 2. User selects the “Export Data” option. 3. System displays the export interface. 4. User selects the students and the data fields to be exported. 5. User selects the export data format and the file destination folder. 6. User initiates the export process. 7. System validates the file export parameters. 8. System performs the data export process. 9. System generates the export file. 10. System provides the download link for the export file or sends the file to the specified destination folder. 11. System displays a message confirming the successful import process of the student data.

Exception Conditions:	• Data matching criteria is not met, if so, then the system notifies the user by prompting an error message. • Certain data fields may have insufficient permissions to export them, if so, the system automatically excludes the restricted data from the export file. • Error occurs while generating the file, if so, then the system will display an error message.

3.1.13 Generate Student Data Report

FR13: The system should allow the system admin and teaching staff to generate student data report

Title:	Generate Student Data Report
Description:	Student Data reports which include personal student information, their medical history, previous school history and contact information of the parents of students are generated based on the student data stored.
Actor(s) Involved:	Teaching Staff, System Administrators
Preconditions:	• The required student data to create the report must be present within the system. • System should have set parameters in place to standardize the student data report. • User should have the required permission to request a form for student data.
Post-Conditions:	• Requested academic reports are generated in the appropriate and specified format. • Reports are available for distribution and download. • System creates a log for the generated academic report.
Triggers:	• User chooses the “Generate Student Data Report” option under the “Reports” section.
Activity Flow:	1. User navigates to the “Reports” section. 2. User selects the report type (Student Data Report). 3. User specifies the output format of the report (PDF, Digital etc). 4. User initiates the report generation process. 5. System fetches and retrieves the required data. 6. System processes the data according to the report template. 7. System generates the student data report document. 8. System provides access to the completed report. 9. System displays a message confirming the successful generation of the student data report.

Exception Conditions:	• Insufficient data present to create a complete report, if this, then the missing data fields are flagged by the system and system prompts an error message.

3.1.14 Enter Student Marks

FR14: The system should allow the teaching staff to enter student marks

Title:	Enter Student Marks
Description:	Teachers record the marks and exam results for students' assessments, assignments, tests and their examinations.
Actor(s) Involved:	Teaching Staff
Preconditions:	• The teacher adding the marks must be logged into the system. • The teacher must have the appropriate permissions to be able to add the examination marks of students. • The students whose marks are being entered must belong to the particular class.
Post-Conditions:	• Student marks are recorded within the system. • System uses the mark entered to calculate automated processes such as class averages and individual grades.
Triggers:	• The user will choose the "Add Marks" option and begin the marks entry process.
Activity Flow:	1. Teacher navigates to the academic mark section. 2. Teacher selected the class, subject and assessment attempted. 3. System displays the student list for the selected class. 4. Teacher then enters the marks for each student. 5. Teacher submits the marks. 6. System validates the marks entered. 7. System records the marks that are scored in a database. 8. System displays a message confirming the successful generation of the academic report.
Exception Conditions.	• Invalid mark value is entered (e.g., 112) which is out of range, if such, the system will prompt the user for value correction. • Student may not be enrolled in the same class, if such, system will display an error message.

3.1.15 View Student Grades

FR15: The system should allow users to view student grades

Title:	View Student Grades
Description:	The system will allow all users to only view the student grade information for any particular student and other details based on the viewer's role and allowed permissions.
Actor(s) Involved:	System Administrators, Teachers, Students, Parents
Preconditions:	• The user has been successfully authenticated into the system. • The user has the required permissions to be able to view the student grades. • The grade data that is to be viewed should be present in the system.
Post-Conditions:	• Grade information is displayed according to the permission level • System creates a log of the grade access request.
Triggers:	• User will select the "View Grades" option to view the grades
Activity Flow:	1. The System Administrator navigates to the View Grades section. 2. The user will fill in special parameters (Student name, Year Level, Class, Year Term) to filter out the required record. 3. System validates the viewing permissions 4. System retrieves the grade details 5. System displays the grade for the required record together with an appropriate level of detail 6. System provides an option for printing or exporting the grades (depends on user role and permissions)
Exception Conditions:	• Insufficient permissions - the system will only display limited information depending on the user role • Missing or incomplete grades - the system will show a message and indicate that data is unavailable.

3.1.16 Update Student Marks

FR16: The system should allow teaching staff to update student marks

Title:	Update Student Marks
Description:	Teachers are assigned the permission to modify previously recorded academic marks when corrections are needed.
Actor(s) Involved:	Teaching Staff
Preconditions:	• The teacher must be authenticated within the system. • Teachers should have the required permissions to edit marks for specific subjects. • The marks to be updated should already be existing in the system.
Post-Conditions:	• Student marks are updated in the system. • System then re-calculates the class and subject averages depending on the changes made. • System creates a log of the marks that have been updated and does it with previous and new values.
Triggers:	• Teacher selects the “Edit Mark” to edit the existing student examination marks.
Activity Flow:	1. Teacher navigates to the academic mark section. 2. Teacher selects the class, subject and the assessment marks. 3. System displays the student list with the current mark. 4. Teacher modifies marks as necessary. 5. Teacher submits the updated marks. 6. System validates the mark entries. 7. The system updates the marks records in the database. 8. System re-calculates values such as class averages and individual grades. 9. System displays a message confirming the successful update of the student mark.

Exception Conditions:	• If an invalid mark value is entered (e.g., 112) which is out of range, if such, the system will prompt the user for value correction.

3.1.17 Delete Student Marks

FR17: The system should allow teaching staff to delete student marks

Title:	Delete Student Marks
Description:	Teachers are assigned the permission to remove and delete incorrectly entered marks or marks for assessments that have been cancelled.
Actor(s) Involved:	Teaching staff
Preconditions:	• The teacher must be authenticated within the system. • Teachers should have the required permissions to delete the marks for specific subjects. • The marks to be deleted should already be existing in the system.
Post-Conditions:	• Student marks will be removed from the active records from where the marks were deleted from. • System re-calculates the affected data metric fields such as class average. • System creates a log of the marks that were deleted.
Triggers:	• Teacher selects the “Delete Marks” option to delete existing mark
Activity Flow:	1. Teacher navigates to the academic mark section. 2. Teacher selects the class, subject and the assessment marks. 3. System displays the student list with the current marks. 4. Teacher selects the marks that are meant to be deleted. 5. System will prompt the user for confirmation. 6. Teacher confirms the mark deletion process. 7. The system updates the marks records in the database and removes the marks. 8. System re-calculates values such as class averages and individual grades. 9. System displays a message confirming the successful deletion of the student marks.

Exception Conditions:	• If reports have already been generated before using the same data, the system will prompt and warn the user of the impact of this on reports. • If multiple related assessments are affected due to deletion of records, the system provides impact analysis.

3.1.18 Generate Academic Report

FR18: The system should allow the system admin and teaching staff to generate academic report

Title:	Generate academic report
Description:	Academic reports which include examination progress reports, report cards and in-class performance summaries all in one are generated based on the student academic data stored.
Actor(s) Involved:	Teaching Staff, System Administrators
Preconditions:	• The required student academic data to create the report must be present within the system. • System should have set parameters in place to standardize the academic report. • User should have the required permission to request an academic report generation cue.
Post-Conditions:	• Requested academic reports are generated in the appropriate and specified format. • Reports are available for distribution and download. • System creates a log for the generated academic report.
Triggers:	Either: • Scheduled report generation (every term-end) OR • User request for an academic report at any time (student changing school).
Activity Flow:	1. User navigates to the “Reports” section. 2. User selects the report type and the report parameters. 3. User specifies the output format of the report (PDF, Digital etc) 4. User initiates the report generation process. 5. System fetches and retrieves the required data. 6. System processes the data according to the report template. 7. System generates the report document. 8. System provides access to the completed report.

	9. System displays a message confirming the successful generation of the academic report.
Exception Conditions:	• Insufficient data present to create a complete report, if this, then the missing data fields are flagged by the system and system prompts an error message.

3.1.19 MyAPA Assessment

FR19: The system should allow the admin staff and teaching staff to conduct MyAPA Assessment on teachers to evaluate teacher performance

Title:	MyAPA Assessment
Description:	MyAPA (Academic Performance Assessment) is a system used to monitor, record and evaluate staff performance metrics, professional development activities and measure performance growth over-time.
Actor(s) Involved:	Administrative Staff and Teaching Staff
Preconditions:	• User must be logged into the system • Administrative staff must be authenticated with the appropriate access permissions to fill in assessment forms for teachers. • The user profiles of the teaching staff being evaluated must exist within the system • Performance evaluation criteria must be pre-defined.
Post-Conditions:	• MyAPA evaluations are recorded and stored in a section of the teaching staffs' profile, which only authorized users can access. • Personal development plans are created and updated. • Teacher performance reports are generated
Triggers:	• The administrative staff does a manual initiation of the teacher MyAPA review initiation. • Upon completion of academic terms (school terms)
Activity Flow:	1. The Administrative staff initiates the performance evaluation process. 2. System notifies all teaching staff about the upcoming staff evaluation 3. Administrative staff perform class observations and complete the MyAPA assessment forms which provide teacher ratings and performance feedback based on the pre-defined criteria. 4. The assessed staff member reviews the evaluation results and provides a self-assessment

	5. System creates a log of the professional activities completed. 6. Teacher performance reports are generated and stored in the user profile of the teaching staff assessed (restricted access)
Exception Conditions:	• If the evaluation period is postponed, teaching staff will be notified.

3.1.20 Manage teacher leave and attendance

FR20: The system should allow admin staff and teaching staff to manage leave and attendance

Title:	Manage teacher leave and attendance
Description:	System will track staff attendance, manage leave requests, keep track of teacher absenteeism and generate summary reports for administrative and staff evaluation purposes.
Actor(s) Involved:	Administrative Staff, Teaching Staff
Preconditions:	• User must be authenticated with the appropriate access permissions to access the summary reports. • Staff profiles that are being monitored must exist within the system.
Post-Conditions:	• Attendance for teachers is updated • Leave requests are processed and logged • Teacher absences are logged, and administrative staff are notified. • Teacher leave balances are adjusted accordingly.
Triggers:	• Daily attendance tracking • Staff member submits a leave request
Activity Flow:	1. Staff member logs into the system to either record daily attendance or submit a leave request. 2. For leave requests: Staff member selects the leave type, dates and the reason for applying for a leave. 3. System checks the leave balance and eligibility 4. Request the administrator for leave approval by selecting the “Apply” option 5. Staff member is then notified upon approval/denial of request. 6. Approved leaves are logged into the personal attendance and leave reports of staff members.

Exception Conditions:	• If there are no more leave balances remaining or staff, the leave request is denied, and the staff member is notified. • For emergency absences without prior notices (e.g., death in family), leaves are permitted regardless of requests.

3.1.21 Record class attendance

FR21: The system should allow the teachers to record class attendance

Title:	Record class attendance
Description:	Teachers can create records of daily class attendance
Actors involved:	Teacher
Preconditions:	• Teachers must be logged into the system. • Teachers should have the necessary permissions to be able to record class attendance. • Student records of a particular class must be present in the system
Post-Conditions:	• Student attendance will be updated. • System logs the attendance record into the database.
Triggers:	• Teacher initiates the attendance recording process for a class session. • End of class period requires attendance submission.
Activity Flow:	1. Teacher logs into the system. 2. Teacher navigates to the attendance module. 3. Teacher selects the specific class/section. 4. System displays the list of enrolled students. 5. Teacher marks each student as present, absent, or late. 6. Teachers can add comments if needed. 7. Teacher submits the attendance record. 8. System validates the data and saves it to the database. 9. System confirms successful recording of attendance.
Exception Conditions:	• If internet connection fails during submission, the system should save data locally and sync when connection is restored. • If a student is not listed in the class roster, the teacher can report the issue to the system administrator.

3.1.22 Record disciplinary incidents

FR22: The system should allow the teachers and the school administrative staff to record disciplinary incidents

Title:	Record disciplinary incidents
Description:	Teachers and administrative staff can record disciplinary incidents regarding students.
Actors involved:	Teachers and administrative staff (Assistant Principal)
Preconditions:	• Teacher and Assistant Principal must be logged into the system. • Teacher and Assistant Principal should have the necessary permissions to be able to record disciplinary incidents. • Student details of the students enrolled in the school must be present
Post-Conditions:	• Disciplinary record of a particular student will be updated • System logs the disciplinary record into the database
Triggers:	• Teacher or administrative staff initiates recording of a disciplinary incident. • Follow-up on previously recorded incidents.
Activity Flow:	1. Teacher or administrative staff logs into the system. 2. User navigates to the disciplinary incidents' module. 3. User selects the "Record New Incident" option. 4. User searches for and selects the student(s) involved. 5. User enters incident details including date, time, location, and description. 6. Users categorise the incident based on severity and type. 7. User adds any relevant witnesses or additional information. 8. User specifies actions taken or consequences applied. 9. User submits the incident report. 10. System validates and saves the information to the database.
Exception Conditions:	• If the student is not found in the system, users can request to add a temporary record pending verification.

	• If multiple students are involved in the same incident, the system should allow linking the same incident to multiple student records. • If a submitted report requires approval, the system should place it in a pending state until reviewed by authorized personnel.

3.1.23 Notify attendance issues

FR23: The teacher should notify parents of attendance issues regarding their child(ren)

Title:	Notify attendance issues
Description:	Teachers can send a notification of attendance issues to parents regarding their child(ren).
Actors involved:	Teacher and Parents
Preconditions:	• Teachers must be logged into the system in order to send the notification. • Parents must be logged into the system in order to receive the notification.
Post-Conditions:	• Parents will be notified regarding attendance issues.
Triggers:	• Student's absence reaches a 4 consecutive day threshold. • Teacher manually initiates attendance notification. • Attendance pattern indicates a potential issue.
Activity Flow:	1. Teacher logs into the system. 2. Teacher navigates to the attendance module. 3. Teachers view students with attendance issues based on system alerts or manual review. 4. Teacher selects the student(s) whose parents need to be notified. 5. System presents a notification template with auto-populated attendance data. 6. Teachers can customize the message if needed. 7. Teacher selects the delivery method (app notification, email, SMS). 8. Teacher sends the notification. 9. System delivers the notification to parents and records the communication.
Exception Conditions:	• If notification delivery fails, the system should retry and notify the teacher of the failure.

3.1.24 Create new class timetable

FR24: The system should allow the system administrator to create new class timetables

Title:	Create new class timetable
Description:	System administrators can create a new timetable for each class.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System Administrator should have the required permissions to create class timetables.
Post-Conditions:	• A new class timetable is created in the system database. • Users are to view the newly created timetable. • The system logs the new timetable creation.
Triggers:	• Beginning of a new academic term or semester. • Addition of new classes or courses. • Major restructuring of existing schedules. • Administrator manually initiates timetable creation.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the timetable management module. 3. Administrator selects the "Create New Timetable" option. 4. Administrator selects the form for which to create the timetable. 5. Administrator specifies the academic term and dates. 6. Administrator adds time slots, subjects, teachers, and rooms to the timetable. 7. System validates entries for conflicts. 8. Administrator reviews the draft timetable. 9. Administrator finalizes and submits the timetable.

	10. System saves the timetable and makes it available to appropriate users.
Exception Conditions:	• If required resources (teachers, rooms) are unavailable, the system alerts the administrator and prevents schedule conflicts. • If any mandatory fields are left blank, the system should prevent submission and highlight the missing information.

3.1.25 Update class timetable

FR25: The system should allow the system admin to update existing class timetables

Title:	Update class timetable
Description:	System administrators can update the timetable for each class.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System Administrator should have the required permissions to update class timetables.
Post-Conditions:	• Class timetable is updated in the system database. • Users are to view the updated timetable. • The system logs the updated timetable.
Triggers:	• Changes in teaching staff or room availability. • Schedule adjustments requested by teachers or administration. • Error correction in existing timetables. • Special events requiring schedule modifications.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the timetable management module. 3. Administrator selects the timetable to update. 4. System displays the current timetable configuration. 5. Administrator makes necessary changes (time slots, subjects, teachers, rooms). 6. System validates changes for conflicts in real-time. 7. Administrator reviews the modified timetable. 8. Administrator saves the changes. 9. System prompts for date of changes and notification options.

	10. System applies updates and notifies affected users if selected.
Exception Conditions:	• If updates create conflicts with other schedules, the system should highlight the conflicts and prevent saving until resolved. • If multiple administrators attempt to modify the same timetable simultaneously, the system should implement locking mechanisms to prevent conflicting changes.

3.1.26 Delete class timetable

FR26: The system should allow the system admin to delete a class timetable

Title:	Delete class timetable
Description:	System administrators can delete the timetable for a class.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System Administrator should have the required permissions to delete class timetables.
Post-Conditions:	• Class timetable is deleted in the system database. • The system logs the deleted timetable.
Triggers:	• End of academic term or year. • Cancellation of classes or courses. • Major restructuring requiring removal of outdated timetables. • Error correction requiring complete removal and recreation.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the timetable management module. 3. Administrator selects the timetable to delete. 4. System displays the timetable details for confirmation. 5. Administrator selects the "Delete Timetable" option. 6. System prompts for confirmation of deletion. 7. Administrator confirms deletion. 8. System checks for dependencies and active references to the timetable. 9. System deletes the timetable from the active database. 10. System confirms successful deletion, logs the action and gives a prompt for notifications.

	11. System applies changes and notifies affected users if selected.
Exception Conditions:	• If the timetable is currently in use (active term), the system should warn the administrator and require additional confirmation. • If the timetable has associated attendance or scheduling data, the system should offer options to archive rather than permanently delete. • If deletion would impact other system functions, the system should identify dependencies before allowing deletion.

3.1.27 Detect scheduling conflicts

FR27: The system should detect scheduling conflicts

Title:	Detect scheduling conflicts
Description:	System detects scheduling conflicts and prevents users from creating a conflicting schedule.
Actors involved:	System administrator, Teachers
Preconditions:	• Existing timetables and scheduling data must be present in the system. • Resource allocation data (rooms, teachers, equipment) must be available.
Post-Conditions:	• Potential conflicts are identified and reported. • System prevents creation of conflicting schedules. • Conflict resolution options are presented to users.
Triggers:	• Creation of new timetables. • Modification of existing schedules. • Resource allocation changes. • Automatic system validation runs.
Activity Flow:	1. System administrators attempt to create or modify a schedule. 2. System automatically checks for the following: a. Teacher assigned to multiple classes at the same time. b. Room allocated to multiple classes at the same time. c. Student groups scheduled for multiple classes simultaneously. 3. If conflicts are detected, the system immediately alerts the administrator. 4. System displays details of each conflict. 5. System suggests possible resolutions when available. 6. Users must resolve all conflicts before saving changes. 7. System validates that the final schedule is conflict-free.

Exception Conditions:	• If emergency overrides are needed, administrators with special privileges can force changes with mandatory documentation. • If conflict involves resources across different departments, the system should notify all relevant administrators. • If temporary conflicts are anticipated due to special events, the system should allow time-limited exceptions with proper documentation.

3.1.28 View class timetable

FR28: The system should allow the system admin, teachers, administrative staff and students to view class timetable

Title:	View class timetable
Description:	Administrator, teachers, administrative staff and students can view class timetable
Actors involved:	Administrator, teachers, administrative staff and students
Preconditions:	• Administrators, teachers, administrative staff and students must be logged into the system.
Post-Conditions:	• Users are to view their respective timetable. • Teachers should have the necessary permissions to be able to record class attendance in extracurricular activities.
Triggers:	• User navigates to the timetable section of the system. • Beginning of new term/semester. • Schedule changes notification prompts viewing. • Daily/weekly routine check of schedules.
Activity Flow:	1. User logs into the system. 2. User navigates to the timetable/schedule section. 3. System identifies user role and access permissions. 4. Based on role, system displays appropriate views: - a. Administrators: Can view all timetables b. Teachers: Can view their teaching schedule and class schedules they teach c. Students: Can view their class schedule only d. Administrative staff: Can view schedules based on their responsibilities 5. Users can select different view formats (daily, weekly, monthly). 6. Users can filter or search for specific classes, teachers, or rooms.
Exception	• If timetable data is being updated, the system should indicate that the displayed information may be changing.

Conditions:	• If the timetable contains placeholder information, the system should clearly mark these entries as provisional.

3.1.29 Record extracurricular activity attendance

FR29: The system should allow the teachers to record student attendance in extracurricular activities

Title:	Record extracurricular activity attendance
Description:	Teachers are able to record student attendance in extra curricular activities.
Actors involved:	Teacher
Preconditions:	• Teachers must be logged into the system. • Student records of a particular class must be present in the system.
Post-Conditions:	• Student attendance in extracurricular activities will be updated. • System logs the attendance in extracurricular activities recorded into the database.
Triggers:	• Scheduled extracurricular activity takes place. • Teacher initiates attendance recording for an activity.
Activity Flow:	1. Teacher logs into the system. 2. Teacher navigates to the extracurricular activities' module. 3. Teacher selects the specific activity from the list. 4. System displays a roster of students enrolled in the activity. 5. Teacher marks each student as present, absent, or excused. 6. Teacher submits the attendance record. 7. System validates the data and saves it to the database. 8. System confirms successful recording of attendance.
Exception Conditions:	• If an unenrolled student attends the activity, the teacher should have the option to add them as a guest participant. • If activity is cancelled, teachers should be able to mark the session as cancelled rather than marking all students absent.

3.1.30 Enrol in extracurricular activities

FR30: The system should allow students to enrol in their respective extracurricular activities

Title:	Enrol in extracurricular activities
Description:	Students are able to enrol in extracurricular activities of their choosing.
Actors involved:	Student
Preconditions:	• Students must be logged into the system. • List of extracurricular activities must be available in the system.
Post-Conditions:	• List of selections is updated. • System logs the updated list in extracurricular activities enrolment into the database.
Triggers:	• Beginning of academic term/semester. • Enrolment period opens for activities. • New activities become available. • Student requests to change activity enrolment.
Activity Flow:	1. Students log into the system. 2. Students navigate to the extracurricular activity's module. 3. System displays available activities based on the student's grade/age eligibility. 4. Students can view details of each activity (schedule, location, instructor, description). 5. Students select activities of interest. 6. System checks for scheduling conflicts and capacity limitations. 7. If no conflicts exist, the student confirms selection. 8. System processes enrolment requests. 9. System updates activity roster and student record.

	10. System confirms successful enrolment and provides activity details.
Exception Conditions:	• If activity has prerequisites or requirements, the system should verify student eligibility before confirming enrolment. • If a student attempts to enrol in activities with scheduling conflicts, the system should prevent enrolment and highlight the conflict.

3.1.31 Create teacher timetables

FR31: The system should allow System Admins to create timetables of teachers

Title:	Create teacher timetables
Description:	System administrators can create timetables for individual teachers showing their teaching schedule.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System Administrator should have the required permissions to create teacher timetables. • Teacher profiles must exist in the system. • Course information must be available in the system.
Post-Conditions:	• New teacher timetable is created in the system database. • Teachers can view their personalized teaching schedule. • The system logs the timetable creation.
Triggers:	• Start of a new academic term/semester. • New teacher joins the school. • Teaching assignments are distributed or modified. • Administrator manually initiates teacher timetable creation.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the teacher timetable management module. 3. Administrator selects the "Create New Teacher Timetable" option. 4. Administrator selects the specific teacher from a list. 5. System displays available time slots and courses. 6. Administrator assigns courses/classes to specific time slots. 7. Administrator specifies classroom locations for each session. 8. System validates entries for conflicts in real-time.

	9. Administrator finalizes and submits the timetable. 10. System saves the timetable and makes it available to the teacher.
Exception Conditions:	• If required courses have no available time slots, the system alerts the administrator. • If teacher workload exceeds contractual limits, the system should display a warning. • If a teacher has specified unavailable times, the system should prevent scheduling during those periods.

3.1.32 Update teacher timetables

FR32: The system should allow System Admin to update timetables of teachers

Title:	Update teacher timetables
Description:	System administrators can modify existing timetables for individual teachers.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System administrators should have the required permissions to edit teacher timetables. • Existing teacher timetables must be present in the system.
Post-Conditions:	• Teacher timetable is updated in the system database. • Teachers can view their updated teaching schedule. • The system logs the timetable modification.
Triggers:	• Changes in teaching assignments. • Scheduling adjustments due to special events or circumstances. • Teacher availability changes. • Error correction in existing timetables. • Room allocation changes.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the teacher timetable management module. 3. Administrator searches for and selects the specific teacher. 4. System displays the current timetable for the selected teacher. 5. Administrator makes necessary changes (time slots, courses, rooms). 6. System validates changes for conflicts in real-time. 7. Administrator saves the changes.

	8. System prompts for notification options. 9. System applies updates and notifies the affected teacher if selected.
Exception Conditions:	• If changes create conflicts with student class schedules, the system should highlight the conflicts and prevent saving until resolved. • If modifications affect other teachers' schedules, the system should alert the administrator and identify the impact. • If changes are made after the term has begun, the system should prompt for additional confirmation and suggest notification methods.

3.1.33 Delete teacher timetables

FR33: The system should allow System Admins to delete timetables of teachers

Title:	Delete teacher timetables
Description:	System administrators can delete timetables for individual teachers when necessary.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System Administrator should have the required permissions to delete teacher timetables. • Existing teacher timetables must be present in the system.
Post-Conditions:	• Teacher timetable is removed from the active system database. • System logs the timetable deletion. • Related dependencies are handled appropriately.
Triggers:	• End of academic term/semester. • Teacher leaves the school. • Complete reassignment of teaching duties. • Major errors requiring fresh creation rather than editing.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the teacher timetable management module. 3. Administrator searches for and selects the specific teacher. 4. System displays the current timetable for the selected teacher. 5. Administrator selects the "Delete Timetable" option. 6. System prompts for confirmation with warning about dependencies. 7. Administrator confirms deletion.

	8. System checks for dependencies (class schedules, room allocations). 9. System deletes the timetable from the active database. 10. System confirms successful deletion and logs the action.
Exception Conditions:	• If the timetable deletion would cause classes with no assigned teacher, the system should prevent deletion and list the affected classes. • If the teacher has active classes with attendance records, the system should offer archival options rather than permanent deletion. • If multiple timetables exist for different terms, the system should confirm which specific timetable(s) to delete.

3.1.34 Create course shells for subjects

FR34: The system should allow System Admins to create course shells for a subject

Title:	Create course shells for subjects
Description:	System administrators can create standardized course shells for subjects that teachers can then customize.
Actors involved:	System administrator
Preconditions:	• System administrator must be logged into the system. • System Administrator should have the required permissions to create course shells. • Subject curriculum standards must be available.
Post-Conditions:	• New course shell is created in the system database. • Teachers can access and customize the course shell. • The system logs the creation of the new course shell.
Triggers:	• New curriculum implementation. • New subject added to school offerings. • Administrator manually initiates course shell creation.
Activity Flow:	1. System administrator logs into the system. 2. Administrator navigates to the course management section. 3. Administrator selects the "Create New Course Shell" option. 4. Administrator enters basic course information (subject name, subject code). 5. Administrator defines the course structure (modules, lessons, forums). 6. Administrator attaches default resources and materials. 7. Administrator sets default assessment components and weights. 8. Administrator submits the completed course shell. 9. System validates and saves the course shell to the database.

Exception Conditions:	• If a course shell already exists for the subject/year level, the system should prompt whether to create a duplicate or modify the existing one. • If course prerequisites are needed, the system should prompt for their definition during creation.

3.1.35 Create Announcement

FR35: The system should allow users to create an announcement

Title:	Create Announcement
Description:	Users can create and broadcast announcements on a school wide or class wide scope. Announcements can consist of text, multimedia and hyperlinks.
Actors Involved:	System administrator, administrative staff, teaching staff.
Preconditions:	• User must be logged into the system.
Post-Conditions:	• Announcement must be successfully saved into the database. • Announcement must be sent to correct user groups and visible to them.
Triggers:	• User clicks the “Create Announcement” button located on their dashboard.
Activity Flow:	1. System displays a form with input fields for announcement title, content and attachments. 2. User enters announcement details. 3. If User is a system administrator or administrative staff, they can optionally specify the scope of the announcement (entire school or for a specific class) 4. User clicks the “Confirm” button. 5. System validates the input and saves the announcement details to the database and the attachment in the object storage bucket. 6. System broadcasts the announcement.
Exception Conditions:	• Attempting to create an announcement without filling in all required fields will display an error message. • Failure while uploading attachments (exceeding file size or unsupported file format) will display an error message.

3.1.36 Update Announcements

FR36: The system should allow users to update an announcement

Title:	Update Announcements
Description:	Users can update an announcement that has already been broadcasted.
Actors Involved:	System administrator, administrative staff, teaching staff.
Preconditions:	• User must be logged into the system. • Announcement being updated must already exist.
Post-Conditions:	• Specified announcement must be successfully updated in the database. • Previous instances of the announcement must be updated on the application and visible to users.
Triggers:	• User clicks the “Edit Announcement” button located on the announcement page.
Activity Flow:	1. System displays a form with input fields for announcement title, content and attachments (which are prefilled with the values of the existing announcement). 2. User updates announcement details. 3. If User is a system administrator or administrative staff, they can optionally edit the scope of the announcement (entire school or for a specific class) 4. User clicks the “Confirm” button. 5. System validates the input and updates the announcement details in the database and the attachment in the object storage bucket.
Exception Conditions:	• Attempting to update an announcement without filling in all required fields will display an error message. • Failure while uploading attachments (exceeding file size or unsupported file format) will display an error message. • Attempting to update an announcement that no longer exists will display an error message.

3.1.37 Delete Announcement

FR37: The system should allow users to delete an announcement

Title:	Delete Announcement
Description:	Users can delete an announcement that has already been broadcasted.
Actors Involved:	System administrator, administrative staff, teaching staff.
Preconditions:	• User must be logged into the system. • Announcement being deleted must already exist.
Post-Conditions:	• Specified announcement must be successfully deleted from the database. • Previous instances of the announcement must be removed from the application and not be visible to users.
Triggers:	• User clicks the “Delete Announcement” button located on the announcement page.
Activity Flow:	1. System displays a confirmation modal. 2. User clicks the “Confirm” button. 3. System will delete the announcement details in the database and the attachment in the object storage bucket.
Exception Conditions:	• Attempting to update an announcement that no longer exists will display an error message.

3.1.38 Share Documents

FR38: The system should allow the system administrator and teachers to share documents

Title:	Share Documents
Description:	Users can upload and share documents to students and parents on a school wide or class wide scope. Documents can be in various formats (e.g. PDF, DOC. PPT, JPG, PNG)
Actors Involved:	Administrative staff, teaching staff
Preconditions:	• User must be logged into the system. • Document to be shared must be of a supported file format.
Post-Conditions:	• System must store the document metadata in the database and the document itself in the object storage bucket. • Document must be visible to all associate users.
Triggers:	• User clicks the “Share Document” button located on their dashboard.
Activity Flow:	1. System displays a form with input fields for document title, description and the file itself. 2. User enters document details and uploads the file. 3. If User is an administrative staff, they can optionally specify the scope of the document (entire school or for a specific class) 4. User clicks the “Confirm” button. 5. System validates the input and saves the document details to the database and the file in the object storage bucket.
Exception Conditions:	• Failure while uploading the document (exceeding file size or unsupported file format) will display an error message.

3.1.39 Direct Messaging

FR39: The system should allow teachers and parents to send direct messages

Title:	Direct Messaging
Description:	Users can send direct messages to one another directly on the platform. Messages can consist of text, multimedia and hyperlinks.
Actors Involved:	Teaching staff, parents
Preconditions:	• User must be logged into the system. • Recipient to whom the message is being sent must exist. • Recipient must approve the initial direct message request.
Post-Conditions:	• Message must be successfully saved into the database. • Message must be sent to the correct recipient and visible to them.
Triggers:	• User clicks the “Send Message” button on the recipient's profile page.
Activity Flow:	1. System displays a chat interface with an input for text messages and an option to add attachments. 2. User enters message body and optionally uploads a file. 3. User clicks the “Send” button. 4. System validates the input and saves the message to the database and the attachment in the object storage bucket. 5. System notifies the recipient about the new message.
Exception Conditions:	• Attempting to send a message without filling in all required fields will display an error message. • Failure while uploading attachments (exceeding file size or unsupported file format) will display an error message.

3.1.40 Reserve Books

FR40: The system should allow users to make reservations for library books

Title:	Reserve Books
Description:	Users can borrow books that are currently available or place a reservation on books that are unavailable. User is then required to return the book before its due date.
Actors Involved:	Administrative staff, teaching staff, students
Preconditions:	• User must be logged into the system. • Book being reserved must exist in the system.
Post-Conditions:	• Book reservation must be successfully saved into the database. • User must be added to the reservation queue for the specified book. • User must be able to view all active reservations on their dashboard. • Notification must be sent to the user once a reserved book becomes available.
Triggers:	• User clicks the “Reserve Book” button in the library and online resources page.
Activity Flow:	1. User searches for the desired book using the search bar, 2. User clicks on the derided book. 3. User clicks the “Create Reservation” button. 4. System saves the reservation to the database.
Exception Conditions:	• Attempting to reserve reference material will display an error message. • Attempting to reserve a book that has already been reserved will display an informational message.

3.1.41 View Due Date

FR41: The system should allow users to view the book due date

Title:	View Due Date
Description:	User can view and track the due dates of books that have been borrowed from the library. This is used to monitor when to return books and those books that are overdue.
Actors Involved:	Administrative staff, teaching staff, students
Preconditions:	• User must be logged into the system. • At least must have been borrowed from the library. • Books' due date that is being tracked must exist in the system.
Post-Conditions:	• User must be able to view a list of all books borrowed, along with their due dates. • User must receive a clear indication of overdue books. • User must receive notification before their book is due.
Triggers:	• User clicks the "View Due Dates" button in the library services page.
Activity Flow:	1. System displays a list of all books that have been borrowed ordered by most recent borrow date. 2. User clicks on a specific book to view additional details. 3. System displays detailed information about the books due date such as how many days till due and actual date.
Exception Conditions:	• Attempting to view the due date of a book that does not exist will display an error message. • Attempting to view due dates if no books have been borrowed will display an informational message.

3.1.42 Issue Fines

FR42: The system issues fines for books that are returned after the due date

Title:	Issue Fines
Description:	System will calculate fines for overdue books and issue them to the borrower. The borrower is required to pay the fines in person during school hours.
Actors Involved:	System, students
Preconditions:	• User must be logged into the system. • Reservation for which the due date fine is being issued must exist in the system. • User who reserved the book must exist in the system.
Post-Conditions:	• Fine must be successfully saved into the database. • Notification must be sent to the user once the fine is calculated.
Triggers:	• System runs a periodic cron job
Activity Flow:	1. System calculates the time remaining for all books that have been borrowed. 2. System calculates the fine if a book is overdue. 3. System sends a notification to the user containing the book details and overdue fine (this is done every time the cron job is run, until the book is returned, and the fine is paid).
Exception Conditions:	• Unavailability of the system (due to failure or maintenance) during the cron job may result in fines not being calculated and issued on time.

3.1.43 Search Resources

FR43: The system should allow users to search for resources

Title:	Search Resources
Description:	User can search for both in house books and digital resources from the library catalogue. Resources include fiction, nonfiction, textbooks, reference materials as well as past exam question papers and solutions.
Actors Involved:	Administrative staff, teaching staff, students
Preconditions:	• User must be logged into the system. • At least one resource item must exist in the system.
Post-Conditions:	• System must display a list of all resources that match the user's search query.
Triggers:	• User clicks on the “Search” icon on the search bar or presses “Enter” while focusing on the search bar.
Activity Flow:	1. System queries library database based on the search terms. 2. System displays a list of all matching resources ordered by semantic relevance.
Exception Conditions:	• Attempting to search for a resource without filling in all required fields will display an error message. • Attempting to search for a resource which does not exist in the system displays an error message.

3.1.44 Download Resources

FR44: The system should allow users to download resources

Title:	Download Resources
Description:	User can download any of the digital resources available in the library services.
Actors Involved:	Administrative staff, teaching staff, students
Preconditions:	• User must be logged into the system. • Resource being downloaded must exist in the system.
Post-Conditions:	• Resources must be successfully downloaded onto the user's device. • User must be able to successfully access the downloaded resource.
Triggers:	• User clicks on the “Download” button on a digital resource.
Activity Flow:	1. System provides download format options to user if any are available. 2. User selects a file format. 3. System downloads the file onto the user’s device.
Exception Conditions:	• Low bandwidth may initiate the download process after a certain duration, giving the impression that the feature does not work.

3.1.45 Chat With AI

FR45: The system should allow users to chat with the integrated AI chatbot

Title:	Chat With AI
Description:	User can chat with an artificial intelligence powered chatbot using natural language. Queries can be based on school rules, upcoming events, academic support, and other relevant topics.
Actors Involved:	Administrative staff, teaching staff, students, parents.
Preconditions:	• User must be logged into the system. • Chatbot must be trained on up-to-date school related information.
Post-Conditions:	• Chatbot must provide a clear relevant response to the user's query.
Triggers:	• User clicks on the “AI Chatbot” icon on their dashboard.
Activity Flow:	1. System applies spelling and grammar correction to the query. 2. System forwards the query to the AI model. 3. The AI model generates a response. 4. System displays the response to the user.
Exception Conditions:	• Attempting to chat without filling in all required fields will display an error message. • Attempting to ask inappropriate questions will display an error message.

3.2 Detailed non-functional requirements

3.2.1 Product Requirements

3.2.1.1 Performance Requirements

- The system must be able to support without experiencing any form of degradation in performance at least:
    - 1,000 concurrent students
    - 10 staff members
    - 2 administrators
- The system must be able to record and process attendance records for a class of 40 students within 5 seconds of data submission.
- The system shall be hosted on AWS EC2 with:
    - 16 virtual CPUs
    - 32 GB RAM
    - 400 GB storage
- Automated features such as student performance and attendance report generation will improve task completion time by reducing manual workload, error handling, and paperwork costs valued at $90,000.
- The system will implement asset compression and lazy loading to ensure:
    - All page's load with minimal load times
    - Support for users with low bandwidth
- The system must be able to process registration and login transactions within 5 seconds and connect a user to their appropriate account.
- The system must be able to load the dashboard and course shell within 4 seconds of registration or login.

3.2.1.2 Dependability Requirements

- The system must implement automatic backups on a weekly basis.
- System availability shall be maintained at 99.9% uptime.
- The system shall be extensively stress tested prior to deployment in user-controlled environments(Docker).
- In the event of a failure, the system shall be restored to the last stable version from backups.
- The system shall complete backup recovery within one hour of catastrophic failure.

3.2.1.3 Security Requirements

- Multi-factor authentication (MFA) and Role-Based Access Control (RBAC) will control access based on user roles.
- Sensitive student information will only be visible to authorized users.

- The system shall enforce HTTPS protocols using SSL (Secure Socket Layer) certificates to ensure encrypted transmission of data between the server and user's device, ensuring confidentiality of sensitive data (login credentials, academic records, personal information).
- The system shall implement password rotations every 90 days for administrators.

3.2.1.4 Usability Requirements

- The user interface will follow the KISS (Keep It Simple, Stupid) principles by ensuring all files, modules and timetables can be accessed within three clicks.
- The system shall be designed to be accessible via any browsers with modern support (Chrome, Safari, Firefox).
- The system shall be responsive across both mobile devices and desktop.
- Use of minimalistic images supports accessibility in guiding the user and avoid overwhelming the user.

3.2.1.5 Technical Requirements

- The system shall implement automated deployment pipelines (CI/CD) using Jenkins for continuous integration and deployment.
- The frontend development shall utilize modular component-based architecture to ensure consistent design and faster development time.
- The system shall automatically adjust assignment due dates according to UTC-12, Fiji time including transitions between daylight savings time.

3.2.1.6 Accessibility Requirements

- All buttons and dropdowns on the system shall be built with focus and hover element functionality to assist keyboard and mouse users.
- All text on the system shall use sans-serif for readability and proper scalability with different browsers and resolutions.
- Use of ARIA (Accessible Rich Internet Application) labels for screen reader users to attach labels to anonymous HTML elements.
- All images, icons and logos will have the alt text attribute to help users with poor vision or screen readers.

3.2.2 Organisational Requirements

6.2.2.1 Environmental Requirements

- The system will be hosted in the cloud (AWS EC2) to eliminate the need for on-premises servers.
- School-wide Wi-Fi and workstation upgrades will ensure full access for staff and students.

3.2.2.2 Operational Requirements

- Ongoing support will be handled by part-time IT administrators ($40,000/year).
- Training material and sessions will be provided for teachers and admin staff during and after deployment
- System maintenance shall be scheduled outside of school hours with at least 72 hours advance notice.

3.2.2.3 Development Requirements

- Built using modern technologies: React.js (frontend), Django/Node.js (backend), and PostgreSQL/AWS DynamoDB (databases).
- System features will be tested through Unit Testing and User Acceptance Testing (UAT) before full rollout.
- Complete documentation and training manuals will be prepared and distributed prior to deployment.
- The system's source code shall follow consistent naming conventions and include inline comments to support future development and maintenance.
- The system will utilize version control using Git to allow backup, branch and merge source code between developers.

3.2.3 External Requirements

3.2.3.1 Regulatory Requirements

- The system will comply with local and international data privacy regulations, including those similar to GDPR (General Data Protection Regulation).
- Parental consent is mandatory for access to student information.

3.2.3.2 Accounting Requirements

- Full cost-benefit analysis conducted:
    - Fixed Costs: $118,000 (development, infrastructure, implementation)
    - Ongoing Costs: $80,000/year

- Annual Benefits: $90,000

- Projected return on investment (ROI) is expected to be $10,000 annually through tangible benefits such as reduced paperwork, errors and increased efficiency.

3.2.3.3 Safety/Security Requirements

- Contingency procedures include paper-based attendance registers and grade sheets for emergency use.
- Scheduled off-peak maintenance and system health checks to prevent unplanned downtime.
- All data related operations will be logged in real-time to prevent data loss.
- Regular vulnerability tests will be conducted bi-annually by third party security experts.

3.2.3.4 Ethical Requirements

- Only authorized personnel (assigned teachers and the student's parent) may access student performance data.
- The system will implement confidentiality disclosure protocols (student privacy and data usage policies) upon user login or registration.

3.2.3.5 Legislative Requirements

- The system will enforce data protection laws and uphold student rights through secure access management and audit trails.
- All usage of personal and academic data will comply with local education and privacy legislation.
- Audit trails will have logs of who accessed or edited records.

3.2.3.6 Localization and Language Requirements

- The system shall use the universal language of English (UK) as the primary language with support for future expansion for Fijian and Hindi translations.

3.2.3.7 Communication and Integration Requirements

- The system shall expose secure RESTful APIs for integration with third-party applications (library system).

Assignment 2 Mark Allocation Sheet

After having discussed it as a group, we recommend the following mark allocation to each group member based on contribution or lack of it throughout the assignment.

Group number: 12 Project Manager: Vishant Kumar

Member Student ID	Percentage Contribution	Responsibilities
S11230430	20%	Question 2.2 to 2.5 & 3.1
S11221203	20%	Question 3.1
S11230987	20%	Question 3.1
S11234883	20%	Question 3.1 to 3.5
S11230686	20%	Question 1 & 2.1

Signed

Member Name	ID	Signature
Vishant Kumar	S11230430	vishant
Anav Chand	S11221203	anav
Aryan Singh	S11230987	aryan
Rohan Nandan	S11234883	rohan
Zhixian Chen	S11230686	zhixian